



A Qualitative Evaluation and Taxonomy of Bar Chart Annotations

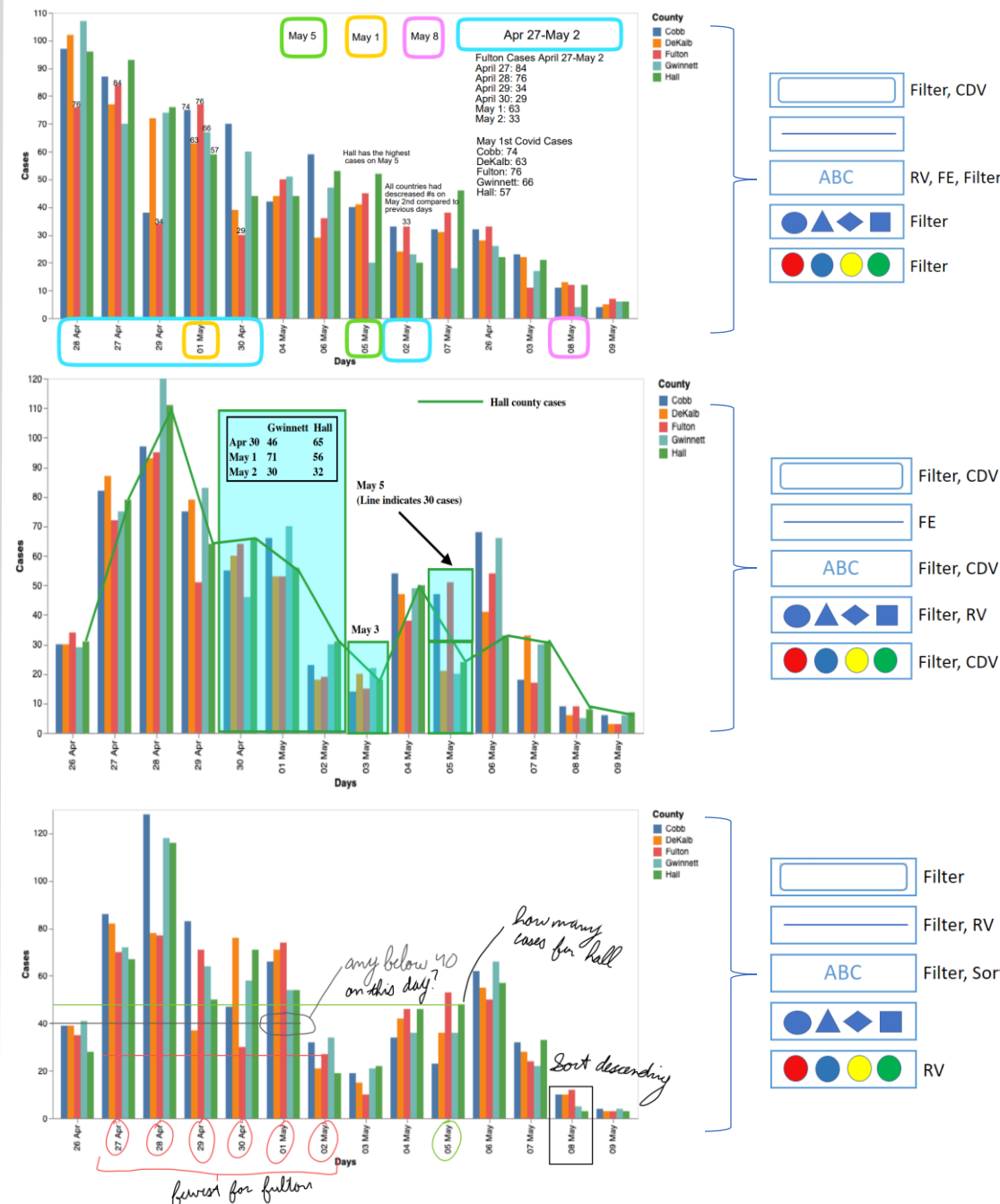
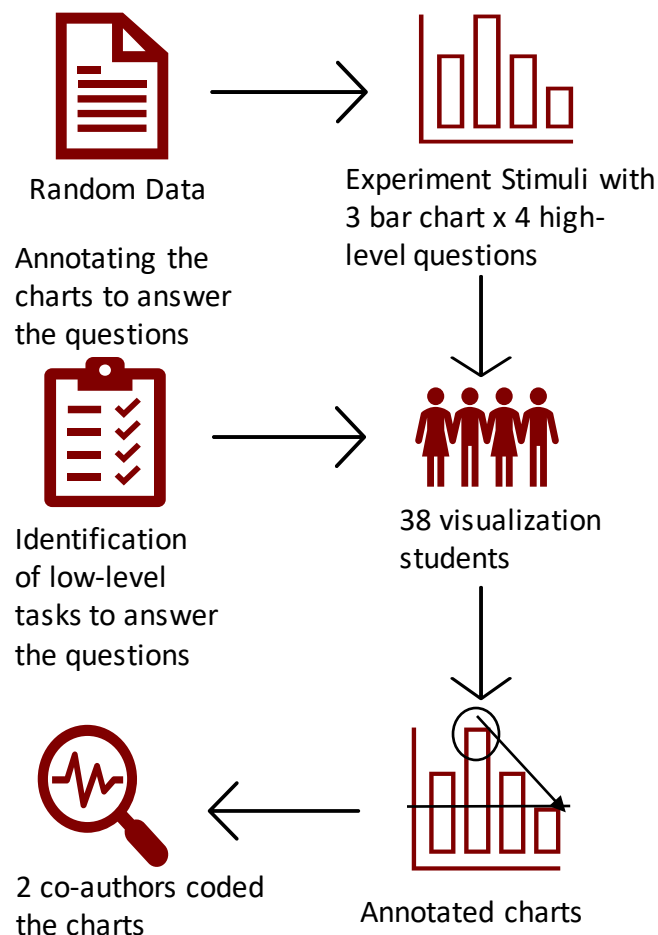
Md Dilshadur Rahman¹ Ghulam Jilani Quadri² Paul Rosen¹
University of Utah¹ University of North Carolina at Chapel Hill²



How do people annotate bar charts?

- We studied how people use annotations in bar charts when they want to communicate information.
- We found interesting usage patterns for annotations and developed a comprehensive taxonomy of bar chart annotations based on those patterns.

Methodology



Participants annotated the charts with different annotation types e.g., Enclosure, Connector, Text, Mark, etc., to denote different low-level tasks e.g., Filter, Retrieve Value (RV), Compute Derived Value (CDV), Find Extremum (FE), and Sort.

Results and Discussion

Task	Enclosure				Connector		Text			Mark		Color			Other	
																
	Ellipse	Bracket	Half Box	Rectangle	Arrow	Line	Description	Values	Legend	Identifier	Legend	Highlights	Category	Questions		Values
Retrieve value	15	4	0	14	18	52	19	11	4	0	1	0	33	14	2	0
Filter	20	18	10	31	34	47	35	4	11	19	10	7	30	55	2	0
Compute Derived Value	1	6	0	8	0	9	11	8	4	0	1	0	15	3	1	5
Find Extremum	11	0	0	5	11	15	21	7	3	0	3	0	14	9	0	0
Sort	0	0	0	1	6	9	14	1	3	0	1	0	6	3	0	0
Total	47	28	10	59	69	132	100	31	25	19	16	7	98	84	5	5

- We identified 15 annotation types grouped into 5 annotation categories.
- Several annotations are used as an ensemble, e.g., mark-color, enclosure-connector-text, etc., to denote a single task.
- The sorting task appeared to be challenging to annotate.
- Participants often used text when they didn't find any suitable annotation type for a given task.

Future Works

- How do people annotate other chart types, e.g., line charts, scatter plots, etc.?
- What are the most effective annotation types for different situations?

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References

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